

Computer Application

Course Description- Information Security

Course Code- 24M11CA119

Subject Code	24M11CA119	Semester: EVEN	Semester II Session 2025 -2026 Month from: JAN to JUNE 2026
Subject Name	Information Security		
Credits	3-0-0	Contact Hours	3Hr.

Faculty (Names)	Coordinator(s)	Dr. Shaili Mishra
	Teacher(s) (Alphabetically)	Dr. Shaili Mishra

COURSE OUTCOMES		COGNITIVE LEVELS
CO1	Discuss the fundamental concepts of information security and malware types	Understand (C2)
CO2	Explain the security risks within an organizational context.	Understand (C2)
CO3	Demonstrate types of cryptographic techniques and working of classical cryptosystems	Apply (C3)
CO4	Determine proactive solutions to network security like Firewalls, access control and IDS	Apply (C3)
CO5	Investigate various security parameters from the perspective of legal and ethical issues	Analyze (C4)

Module No.	Subtitle of the Module	Topics in the Module	No. of Lectures for the module
1.	Introduction to Information Security	Goals of Information Security: Confidentiality, Integrity, and Availability (CIA Triad), Security Terminologies: Threats, Vulnerabilities, Risks, and Attacks	4
2.	Cryptography Concepts and Techniques	Introduction, plain text and cipher text, substitution techniques, transposition techniques, encryption and decryption, symmetric and asymmetric key cryptography, steganography, key range and key size, possible types of attacks.	6

3.	Symmetric and Asymmetric Cryptographic Techniques	DES, AES, Principles of public key cryptosystems, RSA algorithm, Elgamal Cryptography, Diffie-Hellman Key Exchange, Knapsack Algorithm.	6
4.	Authentication and Digital Signatures	Use of Cryptography for authentication, Secure Hash function, Key management – Kerberos	6
5.	Program Security	Non-malicious Program errors – Buffer overflow, Incomplete mediation, Time-of-check to Time-of- use Errors, Viruses, Trapdoors, Salami attack, Man-in-the- middle attacks, Covert channels, , Introduction to Anti-malware technology	8
6.	Security in Networks	Threats in networks, Network Security Controls – Architecture, Encryption, Content Integrity, Strong Authentication, Access Controls, Wireless Security, Traffic flow security, Firewalls – Design and Types of Firewalls, Personal Firewalls, IDS, Email Security – PGP,S/MIME, Access Policies, Implementing Access Control, Procedure-Oriented Access Control, Role-Based Access Control, Captchas	8
7.	Ethical, Legal, and Professional Issues	Cybersecurity Laws and Regulations, Ethical Hacking and Penetration Testing, Privacy and Data Protection, Introduction to Cyber Crimes and Cyber Laws and IT Act 2000	4
Total number of Lectures			42
Evaluation Criteria			
Components		Maximum Marks	
T1		20	
T2		20	
End Semester Examination		35	
TA		25(PBL=10, Assignments =10, Attendance=5)	
Total		100	
Project Based Learning:			
Each student in a group of 3-4 will select a real world application where networking and security concepts are involved. Study the literature around the chosen application. The application will be developed with the use of any open source platform and simulators in its sister lab course. This enhances the student’s knowledge on secured communication applications and helps in enhancing their employability into related sector.			
Recommended Reading material:			
Text Books			
1. "Computer Security: Principles and Practice" by William Stallings and Lawrie Brown			
2. "Cryptography and Network Security" by William Stallings			
3. "Hacking: The Art of Exploitation" by Jon Erickson			
4. Relevant research papers and industry whitepapers			
5. "Blue Team Handbook: Incident Response Edition" by Don Murdoch			
Reference Books			
1. "Security Engineering: A Guide to Building Dependable Distributed Systems" by Ross Anderson			

2. "The Web Application Hacker's Handbook" by Dafydd Stuttard and Marcus Pinto
3. "Practical Malware Analysis" by Michael Sikorski and Andrew Honig
4. "Cybersecurity and Cyberwar: What Everyone Needs to Know" by P.W. Singer and Allan Friedman

Online Resources

1. NIST Special Publications (SP) Series
2. OWASP (Open Web Application Security Project)
3. WE (Common Weakness Enumeration)